

Computer Science & IT

Database Management System

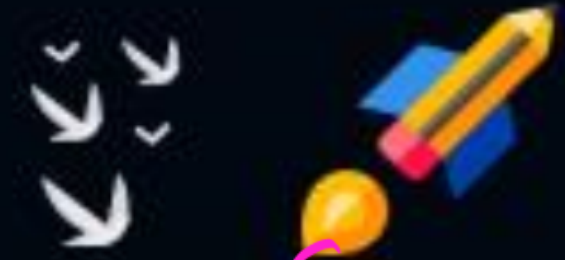


Relational Model & Normal Forms

Lecture No. 09



By- Vishal Sir



Recap of Previous Lecture

- ✓ **Topic** Minimal cover or Canonical cover
- ✓ **Topic** Schema refinement (Normalization)



Topics to be Covered



✓ Topic

Problems because of redundancy in a relation

✓ Topic

Schema refinement (Normalization)

✓ Topic

Properties of decomposition

✓ Topic

Dependency preserving decomposition



Topic : Schema refinement (Normalization)

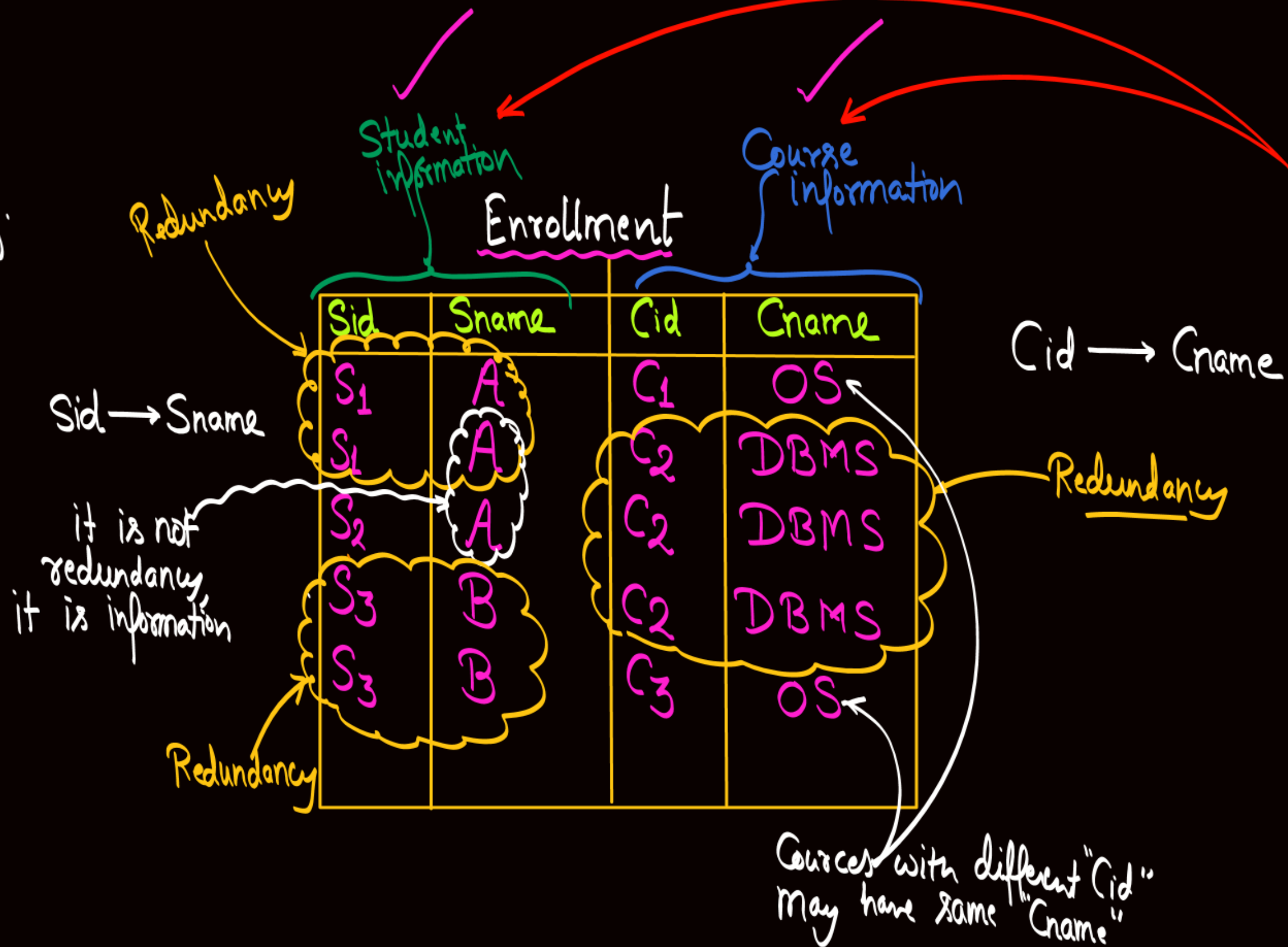
Normalization is a process of reducing / eliminating the redundancy present in the relational table



Topic : Redundancy in a relation

✱ If two or more independent informations are stored in a single table, then redundancy is possible.
(duplicated information)

eg.



We are storing independent informations in the same table.
∴ Redundancy exist. ✓



Topic : Problems because of redundancy

If redundancy is present in database, then various problems are possible

- ① It will require more storage space { Not a very big problem }
- ② Insertion Anomaly
- ③ Deletion Anomaly
- ④ Updation Anomaly

Insertion Anomaly

eg.

Student Information		Course Information	
Enrollment			
Sid	Sname	Cid	Cname
S ₁	A	C ₁	OS
S ₁	A	C ₂	DBMS
S ₂	A	C ₂	DBMS
S ₃	B	C ₂	DBMS
S ₃	B	C ₃	OS
		C ₄	AI

* Sid Can not be NULL,

↳ Until somebody enroll for Course (C₄, AI), we can not insert the information of that course into the table.

If, $Sid \rightarrow Sname$
 $Cid \rightarrow Cname$

are the only two FDs
Present in the relation,

then

Candidate key = (Sid, Cid)

Let this Candidate
key is chosen

as primary key

Then Sid & Cid will not
be allowed to take NULL
Values.

Insertion Anomaly: - If independent informations are stored in the same table, then it may not be possible to insert one independent information without inserting other independent information

eg. (C₄, AI) Can not be inserted,
Until some students enroll for that course

Deletion Anomaly

eg.

If student 'S₁' decides to leave the university then information associated with student 'S₁' must be deleted.

We can not simply update Sid 'S₁' as NULL because Sid is part of Primary key

Student Information		Course Information	
Enrollment			
Sid	Sname	Cid	Cname
S ₁	A	C ₁	OS
S ₁	A	C ₂	DBMS
S ₂	A	C ₂	DBMS
S ₃	B	C ₂	DBMS
S ₃	B	C ₃	OS

If, Sid → Sname
Cid → Cname

are the only two FDs Present in the relation,

then

Candidate key = (Sid, Cid)

Let this Candidate key is Chosen

as primary key

Then Sid & Cid will not be allowed to take NULL Values.

∴ We will have to delete all tuples with Sid = S₁ and because of that we will lose the information of Course with Cid = C₁

Deletion Anomaly: Sometimes it may not be possible to delete one independent information without losing other independent information.

eg. If we delete the tuples w.r.t. $Sid = S_1$, then we also lose the information of Course with $Cid = G_1$
{ Info. of Course "G₂" is still intact because
of other tuples }

Update Anomaly: - Update is required in all duplicate copies, therefore it may be time consuming



Topic : Schema refinement (Normalization)

Normalization is a process of decomposing (splitting) a relational tables into smaller tables (sub-relations) such that it eliminates/reduces the data redundancy, and it can overcome undesirable characteristics like Insertion, Updation and Deletion Anomalies

Student Information

Enrollment

Course Information

Sid	Sname	Cid	Cname
S ₁	A	C ₁	OS
S ₁	A	C ₂	DBMS
S ₂	A	C ₂	DBMS
S ₃	B	C ₂	DBMS
S ₂	B	C ₃	OS

Sid → Sname

Cid → Cname

Assume that relation Enrollment is decomposed into following three relation

Student

Sid	Sname
S ₁	A
S ₂	A
S ₃	B

Sid → Sname

Course

Cid	Cname
C ₁	OS
C ₂	DBMS
C ₃	OS

Cid → Cname

(C₄, AI)

Enrollment_info

Sid	Cid
S ₁	C ₁
S ₁	C ₂
S ₂	C ₂
S ₃	C ₂
S ₂	C ₃

After decomposition there is no redundancy, and we have overcome all the anomalies.



Topic : Properties of decomposition

While decomposing a relation into sub-relations following two properties must be ensured.

① Decomposition must be dependency preserving decomposition
i.e., All functional dependencies present in the original relation must also be preserved even after decomposing the relation into subrelations.

i.e., after decomposition there must not be any loss of information

② Decomposition must be lossless Join decomposition.
i.e., if we perform the natural join ($\bowtie$) operation on all subrelation of original relation, then we must get the exact same set of tuples as of original relation.

(Natural Join)

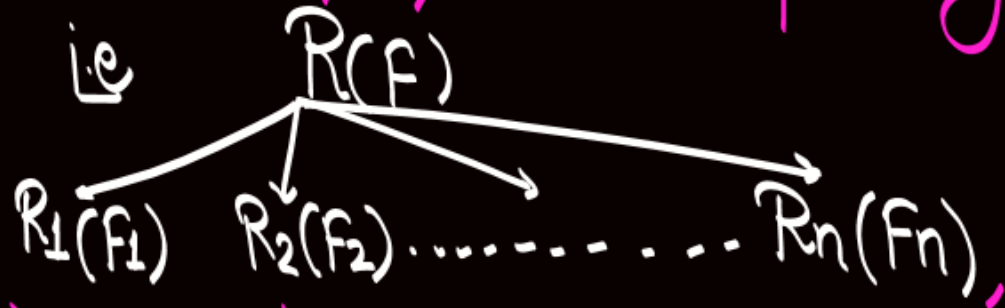
neither in terms of dependency

nor in terms of data

Properties of decomposition

① Dependency Preserving decomposition

Let relation R with FD set F is decomposed into subrelations $R_1, R_2, \dots, R_n$ with FD sets $F_1, F_2, \dots, F_n$ respectively,



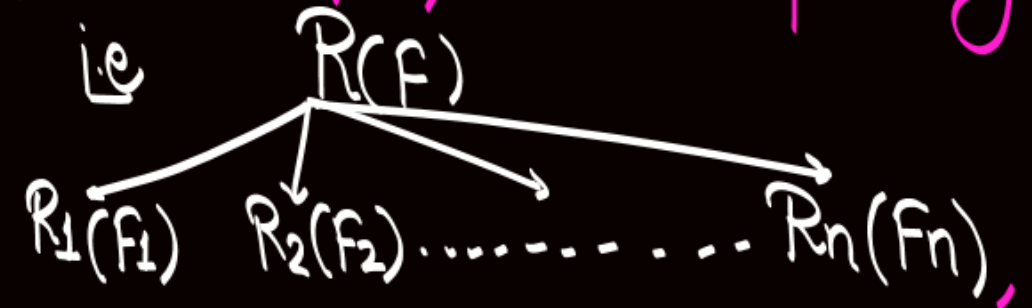
then in general,

$$F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n \subseteq F$$

- (i) if, $F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n = F$, then dependency preserving
- (ii) if, $F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n \subset F$, then not dependency preserving
- (iii) if, $F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n \supset F$, not possible

② Lossless join decomposition

Let relation R with FD set F is decomposed into subrelations $R_1, R_2, \dots, R_n$ with FD sets $F_1, F_2, \dots, F_n$ respectively,



In general,

$$R_1 \bowtie R_2 \bowtie \dots \bowtie R_n \supseteq R$$

if, $= R$, then lossless Join

if, $\supset R$, then Lossy Join

if $\subset R$, never possible



Topic : Dependency preserving decomposition

Q. Let $R(A, B, C, D)$ is a relation with FD set $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$ is decomposed into three subrelations $R_1(A, B)$, $R_2(B, C)$, & $R_3(C, D)$.
Check whether the decomposition is dependency preserving or not?

$R(A, B, C, D)$

$F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

$R_1(A, B)$

$R_2(B, C)$

$R_3(C, D)$

$(A)^+ = \{A, B, C, D\}$
 $(B)^+ = \{A, B, C, D\}$
 $F_1 = \{A \rightarrow B, B \rightarrow A\}$

$(B)^+ = \{B, C, D, A\}$
 $(C)^+ = \{C, D, A, B\}$
 $F_2 = \{B \rightarrow C, C \rightarrow B\}$

$(C)^+ = \{C, D, A, B\}$
 $(D)^+ = \{D, A, B, C\}$
 $F_3 = \{C \rightarrow D, D \rightarrow C\}$

$F_1 \cup F_2 \cup F_3 = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, B \rightarrow A, C \rightarrow B, D \rightarrow C\}$

Check whether $F_1 \cup F_2 \cup F_3 = F$ or not?

Case ① Check if F covers $F_1 \cup F_2 \cup F_3$, i.e., check if $F_1 \cup F_2 \cup F_3 \subseteq F$

Case ② Check if $F_1 \cup F_2 \cup F_3$ covers F , i.e., check if $F \subseteq F_1 \cup F_2 \cup F_3$

We don't need to check for this case, It will always hold true

- FDs of F
- ① $A \rightarrow B$, it is directly available in $F_1 \cup F_2 \cup F_3$
 - ② $B \rightarrow C$ It is directly available in $F_1 \cup F_2 \cup F_3$
 - ③ $C \rightarrow D$ It is directly available in $F_1 \cup F_2 \cup F_3$
 - ④ $D \rightarrow A$, $(D)^+ \text{ w.r.t. } F_1 \cup F_2 \cup F_3 = \{D, C, B, A\}$

$\therefore F \subseteq F_1 \cup F_2 \cup F_3$

$A \in (D)^+ \text{ w.r.t. } F_1 \cup F_2 \cup F_3$
 $\therefore D \rightarrow A$ is a member of $F_1 \cup F_2 \cup F_3$

$F_1 \cup F_2 \cup F_3 \subseteq F$
 $F \subseteq F_1 \cup F_2 \cup F_3$
 $\therefore F_1 \cup F_2 \cup F_3 = F$
 $\therefore$ Dep. preserving decomposition

① H.W.:-

Let $R(A, B, C, D)$ is a relation with FD set,

$$F = \{ AB \rightarrow CD, D \rightarrow A \},$$

If relation R is decomposed into two subrelations

$R_1(AD)$ & $R_2(BCD)$, then Check whether the decomposition is dependency preserving or not.

② H.W.

Let $R(A, B, C, D, E, F)$ is a relation with FD set
 $F = \{ AB \rightarrow C, BC \rightarrow A, AC \rightarrow B, \\ B \rightarrow D, AD \rightarrow E, E \rightarrow F \}$

Which of the following decomposition is dependency
Preserving decomposition

(i) $\mathcal{D}_1 = \{ R_1(A, B, C), R_2(A, C, D), R_3(A, C, E, F) \}$

(ii) $\mathcal{D}_2 = \{ R_1(A, B, C), R_2(E, F), R_3(A, B, D, E) \}$



2 mins Summary

✓
Topic

Problems because of redundancy in a relation

✓
Topic

Schema refinement (Normalization)

✓
Topic

Properties of decomposition

✓
Topic

Dependency preserving decomposition

Tomorrow's Topic: Lossless Join Decomposition 

Home Work:

Read about following relational algebra operations.

- ① Selection (σ)
- ② Projection (π)
- ③ Cross product ($\times$)



THANK - YOU